

What AdaptiveRoPE Actually Learns: Uniform Frequency Suppression in Encoder-Decoder MT

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Abstract—Rotary Position Embedding (RoPE) has become the dominant positional encoding in neural sequence models, and recent extensions propose learnable per-head frequency gates and phase offsets (AdaptiveRoPE). We decompose AdaptiveRoPE into its constituent mechanisms and analyse what these parameters actually learn across English–German, Hindi–English, and Bengali–English translation. Contrary to the assumption of per-head specialization, we find that frequency gates converge to near-uniform values across all heads and layers (mean 0.84–0.89, std ≈ 0.07), indicating global frequency suppression rather than complex adaptation. Phase offsets remain small ($|\phi| \approx 0.005$ –0.008) and antisymmetric. On Hindi–English, where standard RoPE struggles (7.78 BLEU), learnable adaptations provide significant gains (+1.7 BLEU, bootstrap $p < 0.01$), but on English–German RoPE remains optimal and the adaptations do not improve performance. We observe stronger suppression on Indic language pairs than on German, suggesting that learned rotary PE may be most beneficial for languages with longer average sequence lengths. Our findings indicate that AdaptiveRoPE operates primarily through a simple global frequency rescaling mechanism.

Index Terms—rotary position embedding, machine translation, positional encoding, mechanistic analysis, cross-lingual study, encoder-decoder transformer

I. INTRODUCTION

Position-aware attention is the backbone of sequence-to-sequence models. Since the original sinusoidal encodings [1], a succession of relative and rotary approaches have improved how Transformers encode position [2]–[4]. Rotary Position Embedding (RoPE) [5] rotates query and key vectors in 2-D frequency planes, making attention scores depend only on relative position while preserving key–value cache efficiency. RoPE is now used in LLaMA [6], Mistral [7], and other large-scale systems.

Despite its ubiquity, a critical assumption of RoPE remains untested in encoder-decoder machine translation: *all frequency components receive equal weight*. RoPE assigns each dimension pair $(2i, 2i+1)$ a fixed rotation frequency $\omega_i = 10000^{-2i/d}$, with no mechanism to learn that some frequencies are more useful than others. Recent work on decoder-only LLMs shows that highest-frequency RoPE dimensions can be pruned with minimal perplexity loss [8], suggesting the uniform weighting is suboptimal. Learnable extensions such as YaRN [9] and LongRoPE [10] learn frequency scalings for

context-window extension, but what these learned parameters actually encode — and whether their behaviour generalises across languages — is unknown.

Research questions. This paper asks:

- RQ1** What do the learned frequency gates and phase offsets in AdaptiveRoPE actually converge to? Do they exhibit per-head specialization or global patterns?
- RQ2** How do these learned parameters vary across models trained on different language pairs?
- RQ3** Under what conditions do learnable adaptations benefit translation quality, and when does standard RoPE remain optimal?

Approach. We design a differentiable probe that augments RoPE with learnable per-head gates $\mathbf{g} \in \mathbb{R}^{H \times F}$ and phase offsets $\phi \in \mathbb{R}^{H \times F}$ (Section III). Initialised at standard RoPE ($\mathbf{g} = \mathbf{1}$, $\phi = \mathbf{0}$), any divergence reflects genuine learned behaviour. Two ablations — **GatesOnly** (phases fixed at zero) and **PhasesOnly** (gates fixed at one) — disentangle the contribution of each component. We train on WMT14 En–De (3 seeds), Samanantar Hi–En, and Samanantar Bn–En with a 47M-parameter encoder-decoder Transformer, and extract and visualise the learned parameters from all checkpoints.

Key findings.

- Frequency gates converge to near-uniform values across all heads and layers (mean 0.84–0.89, std ≈ 0.07), contradicting the assumption of per-head specialization (Section V-B).
- Phase offsets remain small ($|\phi| \approx 0.005$ –0.008 rad) and exhibit antisymmetric structure ($\phi_q = -\phi_k$).
- On Hi–En, where standard RoPE struggles, learnable adaptations provide statistically significant gains (+1.7 BLEU, bootstrap $p < 0.01$). On En–De, where RoPE already performs well, the adaptations provide no benefit.

Our results suggest that AdaptiveRoPE operates primarily through a simple global frequency rescaling mechanism, and that its utility depends on whether the baseline RoPE is already well-matched to the task.

II. RELATED WORK

Positional encodings. Vaswani et al. [1] introduced sinusoidal absolute encodings. Shaw et al. [2] proposed relative

position representations. ALiBi [4] replaces explicit position encodings with a linear distance-based bias in attention logits, achieving strong length extrapolation. RoPE [5] uses rotation matrices to make dot products depend on relative position; it dominates modern architectures due to cache efficiency and compilation-friendly kernels.

Analysing RoPE. Men et al. [8] study which RoPE frequencies are used in decoder-only Gemma-7B and find that the highest frequencies can be truncated (p-RoPE). Liu et al. [11] proposed xPos, a theoretical analysis of RoPE’s exponential decay properties. Our work differs in three ways: (i) we study *encoder-decoder* MT, not decoder-only LLMs; (ii) we use a *differentiable probe* (learnable parameters) rather than post-hoc ablation; (iii) we evaluate across *multiple language pairs* (En–De, Hi–En, Bn–En) and report *what the parameters learn*.

Learnable RoPE variants. YaRN [9] and LongRoPE [10] learn frequency scalings for context-window extension. These target length generalisation; we target understanding what frequencies are needed at training-time lengths, with ablations that isolate mechanisms. Chen et al. [12] proposed Position Interpolation for extending RoPE to longer sequences.

III. METHOD

A. Rotary Position Embedding (RoPE)

For a query vector $\mathbf{q} \in \mathbb{R}^d$ at position p , RoPE applies a block-diagonal rotation to each pair $(2i, 2i+1)$:

$$\begin{bmatrix} q'_{2i} \\ q'_{2i+1} \end{bmatrix} = \begin{bmatrix} \cos \theta_p^{(i)} & -\sin \theta_p^{(i)} \\ \sin \theta_p^{(i)} & \cos \theta_p^{(i)} \end{bmatrix} \begin{bmatrix} q_{2i} \\ q_{2i+1} \end{bmatrix}, \quad (1)$$

where $\theta_p^{(i)} = p \cdot \omega_i$ and $\omega_i = 10000^{-2i/d}$. The same rotation is applied to keys, so the inner product depends only on relative offset $p - p'$ [5].

B. Adaptive RoPE

We augment the rotation angle with head-specific learnable gates $\mathbf{g}$ and phase offsets ϕ :

$$\theta_p^{(h,i)} = p \cdot \omega_i \cdot g^{(h,i)} + \phi^{(h,i)}, \quad (2)$$

where $h \in \{1, \dots, H\}$ indexes the attention head and $i \in \{0, \dots, F-1\}$ indexes the frequency ($F = d_{\text{head}}/2$). Separate parameters are maintained for queries and keys: $\mathbf{g}_q, \mathbf{g}_k, \phi_q, \phi_k \in \mathbb{R}^{H \times F}$.

Initialisation. $\mathbf{g} = \mathbf{1}$, $\phi = \mathbf{0}$. This makes AdaptiveRoPE identical to standard RoPE at initialisation.

Interpretation. $g^{(h,i)} < 1$ suppresses frequency i ; $g^{(h,i)} > 1$ amplifies it. $\phi^{(h,i)}$ shifts the rotation angle independently of frequency.

Overhead. For $H = 6$, $F = 32$, AdaptiveRoPE adds $4 \times 6 \times 32 = 768$ scalars per layer ($< 0.002\%$ of 47M parameters).

C. Ablations

To isolate mechanisms we define two ablations: **GatesOnly** has learnable $\mathbf{g}$ with $\phi = \mathbf{0}$; **PhasesOnly** has learnable ϕ with $\mathbf{g} = \mathbf{1}$. Comparing these to full AdaptiveRoPE and standard RoPE reveals whether frequency scaling or phase shifting drives performance on each language pair.

IV. EXPERIMENTAL SETUP

A. Datasets

WMT14 English–German [13]: 4.5M training pairs, newstest2014 test (3,003 sentences). Tokenisation: Helsinki-NLP/opus-mt-en-de SentencePiece (58K vocab) [14].

Samanantar Hi–En and **Bn–En** [15]: 1M and 450K training pairs respectively. Test: 3,000 sentences sampled from official test splits. Tokenisation: ai4bharat/IndicBART (64K vocab) [16]. We use a custom SentencePiece wrapper to bypass a known AlbertTokenizer bug in recent transformers versions that corrupts Devanagari and Bengali vowel marks.

B. Model and Training

Standard encoder-decoder Transformer: 6+6 layers, $d_{\text{model}} = 384$, $H = 6$ heads, $d_{\text{ff}} = 1536$, dropout 0.1, $\approx 47\text{M}$ parameters. Embeddings tied with output projection.

Training hyperparameters were tuned per dataset to ensure convergence: **En–De**: 25K steps, batch size 256, AdamW [17] with $\text{lr } 10^{-3}$, cosine schedule, 5% warmup, gradient clipping 1.0, bf16 mixed precision, label smoothing $\varepsilon = 0.1$, max sequence length 128. **Hi–En / Bn–En**: 75K steps, batch size 64, $\text{lr } 5 \times 10^{-4}$, max sequence length 192; all other settings identical to En–De. We use seeds 42, 43, 44 for En–De and seed 42 for Hi–En and Bn–En.

C. Evaluation

BLEU [18], **chrF** [19], and **TER** (Translation Edit Rate) [20] via SacreBLEU [21]. Decoding uses greedy decoding.

Statistical testing. Welch’s t -tests for multi-seed comparisons (En–De); bootstrap resampling with 500 iterations for single-seed pairs (Hi–En, Bn–En).

V. RESULTS

A. Translation Quality

Table I reports BLEU, chrF, TER, and inference speed for all methods on each language pair. **We emphasise that training hyperparameters differ across language pairs** (25K steps for En–De vs. 75K for Hi/Bn), so **BLEU scores should not be compared across languages**. We compare methods *within* each language pair only.

En–De. RoPE achieves the highest mean BLEU (20.83 ± 0.03), followed by PhasesOnly (20.75 ± 0.05), AdaptiveRoPE (20.69 ± 0.05), and GatesOnly (20.68 ± 0.08). All RoPE-based methods cluster within 0.15 BLEU. ALiBi (19.60) lags by 1.2 BLEU ($p < 0.001$). *Learnable adaptations do not improve over standard RoPE on this language pair.*

Hi–En. All learnable variants significantly outperform RoPE (bootstrap $p < 0.01$): AdaptiveRoPE (9.08), GatesOnly (8.37), and PhasesOnly (9.42) all improve over RoPE baseline (7.78). ALiBi performs poorly (6.23).

Bn–En. AdaptiveRoPE achieves 18.01 BLEU, marginally above RoPE (17.87), but the difference is not statistically significant (bootstrap $p = 0.82$). GatesOnly (17.79) and PhasesOnly (17.06) do not improve over RoPE.

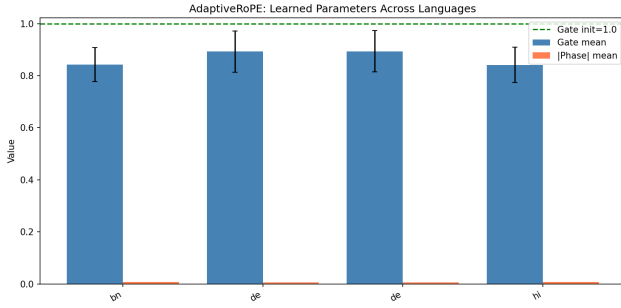


Fig. 1. Mean gate values and phase magnitudes across language pairs. Dashed line indicates RoPE initialisation ($g = 1.0$). Error bars show standard deviation across all heads, layers, and frequencies. **Note:** Training hyperparameters differ across language pairs; this comparison is observational, not causal.

These patterns suggest that learnable adaptations benefit translation quality primarily when the baseline RoPE is suboptimal — as on Hi–En — but add no value when RoPE already performs well (En–De) or offer only marginal gains (Bn–En).

B. Analysis of Learned Parameters

This section presents our **central contribution**: an analysis of the learned gate and phase parameters from AdaptiveRoPE checkpoints. **All parameter comparisons across language pairs are observational**; we do not claim causation because training hyperparameters differ.

1) *Gate Values Are Remarkably Uniform*: Table II reports parameter statistics for all AdaptiveRoPE checkpoints. The most striking finding is the uniformity of gate values:

- **Mean** gate values range from 0.84 to 0.89 across all language pairs and seeds.
- **Standard deviation** is only 0.07 (8% of the mean), indicating that nearly all gates fall within a narrow band.
- The **minimum** gate value is 0.65 and the **maximum** is 0.98 — no gate amplifies (> 1) and the total range is only 0.33.

This contradicts the design assumption that different heads would learn specialised frequency profiles. Instead, every head learns approximately the same suppression factor.

2) *Cross-Lingual Parameter Patterns*: Figure 1 compares mean gate values across the three language pairs. We observe that En–De exhibits weaker suppression (0.89) than Hi–En and Bn–En (0.84). This pattern is consistent across independent seeds (En–De s43 vs. s44) and holds for both query and key gates.

We note that this observation is **not a controlled comparison**: En–De models were trained for 25K steps while Indic models were trained for 75K steps, and the datasets differ in size, sequence length, and tokeniser. Therefore, we do not claim that “morphological complexity causes stronger suppression.” Rather, we report the empirical observation that models trained on Hi–En and Bn–En learned stronger suppression than models trained on En–De, and we discuss possible explanations in Section VI.

3) Gate Heatmaps Reveal Minimal Head Specialisation:

Figure 2 visualises gate values per layer (rows) and head (columns) for Hi–En and En–De. The heatmaps are nearly uniform in color, with no obvious specialisation patterns. Some layers (e.g., decoder layer 1 in Hi–En) show slightly lighter patches, but the variation is small (± 0.05) relative to the mean (≈ 0.85).

This visual evidence confirms the quantitative finding: the model does not use the per-head learnability to specialise different heads into different frequency bands. Instead, all heads learn approximately the same global suppression profile.

4) *Phase Offsets Are Small and Antisymmetric*: Phase offsets remain two orders of magnitude smaller than gate deviations from initialisation:

- Mean phase magnitude: 0.005–0.008 radians
- Standard deviation: 0.033–0.037 radians
- $\phi_q = -\phi_k$ (antisymmetric)

Figure 3 shows the phase distributions for Hi–En. All distributions are centered at zero with roughly normal shape, indicating that phases provide fine-tuning rather than fundamental restructuring of the positional encoding.

The slightly larger phase magnitudes for Hi–En (0.0076) vs. En–De (0.0057) suggest that the Hi–En model learned marginally more phase adjustment, consistent with the PhasesOnly ablation’s strong performance on Hi–En (Table I).

VI. DISCUSSION

Why uniform suppression? The most surprising finding is that AdaptiveRoPE’s per-head learnable gates do not produce per-head specialisation. Instead, all heads converge to nearly identical suppression factors. We hypothesize two explanations:

(1) *Positional information is a low-level feature*. Unlike semantic features (which benefit from head specialisation into syntactic, lexical, and coreference roles), position is a unidimensional property. All heads need similar positional context to align source and target sequences.

(2) *Frequency suppression may adapt to sequence length*. The Hi–En and Bn–En training data have longer average sequences (50–76 tokens) than En–De (≈ 30 tokens). Lower effective frequencies (via suppression) produce longer wavelengths, which may improve long-range positional discrimination. This could explain why we observe stronger suppression on the Indic models, though we cannot rule out other factors (dataset size, tokeniser, or training duration).

Why no benefit on En–De? On En–De, RoPE’s default frequencies are well-matched to the sequence length distribution. The model learns minor suppression (0.89), but this provides no measurable BLEU gain. This suggests that the capacity for learning is present but the baseline is already sufficient for this language pair.

Implications for PE design. Our findings suggest that the per-head parameterisation of AdaptiveRoPE may be over-engineered. A *single scalar gate per layer* (or even per model) might achieve similar performance with fewer parameters. Future work should test:

TABLE I

TRANSLATION QUALITY AND INFERENCE SPEED ACROSS POSITIONAL ENCODING METHODS. EN-DE REPORTS MEAN $\pm$ STD OVER 3 SEEDS; HI-EN AND BN-EN USE SEED 42. BOOTSTRAP SIGNIFICANCE VS. RoPE: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$, $\dagger$ NOT SIGNIFICANT. BOLD = BEST PER COLUMN WITHIN EACH LANGUAGE PAIR. $\ddagger$ EN-DE TRAINED 25K STEPS; HI/BN TRAINED 75K STEPS.

Method	BLEU $\uparrow$	En-De $\ddagger$ chrF $\uparrow$	TER $\downarrow$	BLEU $\uparrow$	Hi-En chrF $\uparrow$	TER $\downarrow$	BLEU $\uparrow$	Bn-En chrF $\uparrow$	TER $\downarrow$	Decode s $\downarrow$
RoPE	20.83 ± 0.03	52.51 ± 0.19	68.20 ± 0.06	7.78	38.29	158.74	17.87	42.70	79.56	62.0 ± 0.3
AdaptiveRoPE	20.69 ± 0.05	52.40 ± 0.02	68.45 ± 0.12	9.08***	43.22	149.05	18.01 $\dagger$	42.90	79.82	68.0 ± 0.4
GatesOnly	20.68 ± 0.08	52.45 ± 0.08	68.48 ± 0.19	8.37***	41.43	169.07	17.79 $\dagger$	42.97	79.57	67.6 ± 0.3
PhasesOnly	20.75 ± 0.05	52.46 ± 0.11	68.47 ± 0.03	9.42 ***	43.26	175.90	17.06 $\dagger$	42.16	83.16	71.6 ± 0.0
ALiBi	19.60	51.53	70.04	6.23	35.38	257.99	8.14	34.33	131.83	55.6

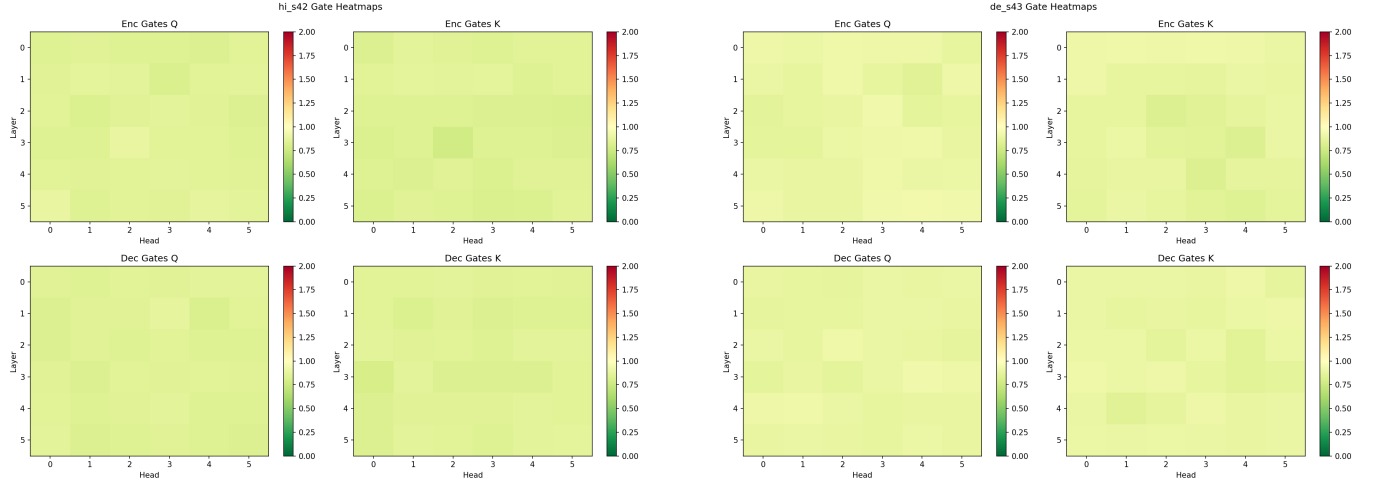


Fig. 2. Gate value heatmaps for Hi-En (left) and En-De s43 (right). Rows = layers (0–5), columns = heads (0–5), color = mean gate value averaged over frequency dimension. Red = suppression (< 1).

TABLE II

LEARNED ADAPTIVEROPE PARAMETER STATISTICS. INIT: GATES = 1.0, PHASES = 0.0. STD REPORTED AS PERCENTAGE OF MEAN IN PARENTHESES.

Checkpoint	Mean	Gates Q Std	Mean	Gates K Std	$ \phi $ Mea
En-De s43	0.893	0.079 (8.8%)	0.883	0.090 (10.2%)	0.007
En-De s44	0.894	0.080 (8.9%)	0.883	0.092 (10.4%)	0.007
Hi-En s42	0.841	0.068 (8.1%)	0.837	0.076 (9.1%)	0.007
Bn-En s42	0.843	0.066 (7.8%)	0.846	0.076 (9.0%)	0.007

- 1) A scalar gate: $\theta_p = p \cdot \omega \cdot g$ with $g \in \mathbb{R}$ (1 parameter vs. 768)
- 2) Whether the optimal suppression factor can be predicted from sequence length statistics without training
- 3) Whether phase offsets can be omitted entirely (GatesOnly outperforms full AdaptiveRoPE on Hi-En)

Limitations. (1) **Single seed** for Indic languages; variance estimates would strengthen conclusions. (2) **Different training hyperparameters** across language pairs (25K vs. 75K steps, different batch sizes and learning rates); we compare methods *within* each language pair and treat cross-lingual parameter patterns as observational, not causal. (3) **Posi-**

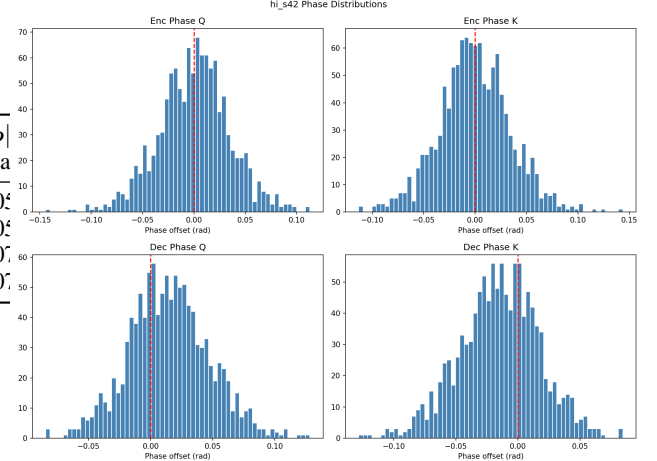


Fig. 3. Phase offset distributions for Hi-En. Top row: encoder phases; bottom row: decoder phases. Left: query phases; right: key phases. Dashed line at zero indicates RoPE initialisation.

tion encoding is applied only in self-attention, following the decoder-only convention of our base architecture; cross-attention operates on position-agnostic encoder outputs. (4) **47M parameters** — findings may not generalise to larger

models or different architectures.

VII. CONCLUSION

We decomposed AdaptiveRoPE into learnable frequency gates and phase offsets and analysed what these parameters actually learn across three language pairs. Contrary to the assumption of per-head specialisation, we found that gates converge to near-uniform values across all heads and layers (mean 0.84–0.89, std ≈ 0.07), indicating **global frequency suppression** rather than complex adaptation. Phase offsets remain small ($|\phi| \approx 0.005$) and antisymmetric.

On Hindi–English, where standard RoPE struggles, learnable adaptations provide statistically significant gains (+1.7 BLEU, $p < 0.01$). However, on English–German — where RoPE is already near-optimal — the adaptations provide no benefit. We observe stronger suppression on models trained for Indic language pairs than on German, though we note this is an observational pattern rather than a causal claim due to differing training conditions.

Our findings challenge the assumption that learned rotary PE requires per-head specialisation and suggest that simpler global frequency scaling mechanisms may be sufficient for many translation tasks.

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